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10 CFR 50.73

July 31, 2019

Serial: RA-19-0292

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555

Subject: Brunswick Steam Electric Plant, Unit No. 2
Renewed Facility Operating License No. DPR-62
Docket No. 50-324
Licensee Event Report 2-2019-003

In accordance with the Code of Federal Regulations, Title 10, Part 50.73, Duke Energy Progress, LLC, submits the enclosed Licensee Event Report (LER). This report fulfills the requirement for a written report within sixty (60) days of a reportable occurrence.

This document contains no regulatory commitments.

Please refer any questions regarding this submittal to Mr. Stephen Yodersmith, Brunswick – Regulatory Affairs, at (910) 832-2568.

Sincerely,

A handwritten signature in blue ink, appearing to read "WRG", is written over a faint, larger blue ink signature.

William R. Gideon

SBY/sby

Enclosure: Licensee Event Report

cc (with enclosure):

U. S. Nuclear Regulatory Commission, Region II
ATTN: Ms. Laura Dudes, Regional Administrator
245 Peachtree Center Ave, NE, Suite 1200
Atlanta, GA 30303-1257

U. S. Nuclear Regulatory Commission
ATTN: Mr. Gale Smith, NRC Senior Resident Inspector
8470 River Road
Southport, NC 28461-8869

U. S. Nuclear Regulatory Commission
ATTN: Mr. Dennis J. Galvin
11555 Rockville Pike
Rockville, MD 20852-2738

Chair - North Carolina Utilities Commission **(Electronic Copy Only)**
4325 Mail Service Center
Raleigh, NC 27699-4300
swatson@ncuc.net

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name

Brunswick Steam Electric Plant (BSEP), Unit 2

2. Docket Number

05000324

3. Page

1 OF 3

4. Title

High Pressure Coolant Injection (HPCI) Inoperable

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
06	12	2019	2019	- 003	- 00	07	31	2019	Facility Name	Docket Number
										05000
										05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER**Licensee Contact**

Stephen Yodersmith, Brunswick – Regulatory Affairs

Telephone Number (Include Area Code)

(910) 832-2568

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
X	BJ	65	W290	Yes					

14. Supplemental Report Expected☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No**15. Expected Submission Date**

Month Day Year

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

At 21:27 Eastern Daylight Time (EDT) on June 12, 2019, during routine testing on Unit 2, the High Pressure Coolant Injection (HPCI) turbine experienced an overspeed trip. Following the overspeed trip, the turbine coasted down and reset, then restarted and rolled to rated conditions as-designed. As a result, the response time of the system exceeded the 60-second acceptance criteria, thereby rendering the system inoperable. Reactor Core Isolation Cooling (RCIC) System and Automatic Depressurization System (ADS) remained operable during this event.

Failure of Unit 2 HPCI to meet the 60 second response time acceptance criteria was caused by a momentary failure of the Electric Governor – Remote (EG-R) to control HPCI Turbine speed during the initial turbine roll. The EG-R Actuator was replaced and the testing was repeated successfully with the response time acceptance criteria met. Historical data was reviewed on Unit 1 and it was confirmed that this condition was isolated to Unit 2.

The safety significance of this event is minimal. There was no impact on the health and safety of the public or plant personnel. This event is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D).

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Brunswick Steam Electric Plant (BSEP), Unit 2	05000324	2019	- 003	- 00

NARRATIVE

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].

Background*Initial Conditions*

At the time of the event, Unit 2 was in Mode 1 (i.e., Power Operation), at approximately 100 percent of rated thermal power (RTP). There was no inoperable equipment that contributed to the event.

Reportability Criteria

The High Pressure Coolant Injection (HPCI) [BJ] inoperability is being reported in accordance with 10 CFR 50.73(a)(2)(v)(D) as a condition that could have prevented the fulfillment of the function needed to mitigate the consequences of an accident. The event was initially reported to the NRC on June 13, 2019 (i.e., Event Number 54116).

Event Description

At 21:27 Eastern Daylight Time (EDT) on June 12, 2019, during routine testing, the Unit 2 High Pressure Coolant Injection (HPCI) turbine experienced an overspeed trip. Following the overspeed trip, the turbine coasted down and reset, then restarted and rolled to rated conditions as-designed. As a result, the response time of the system exceeded the 60-second acceptance criteria, thereby rendering the system inoperable.

Event Cause

Failure of Unit 2 HPCI to meet the 60 second response time acceptance criteria was caused by a momentary failure of the Electric Governor – Remote (EG-R) to control HPCI Turbine speed during the initial turbine roll.

Safety Assessment

There was no adverse impact on the health and safety of the public. The safety significance of this event is minimal. The event occurred during testing, and Reactor Core Isolation Cooling (RCIC) System and Automatic Depressurization System (ADS) remained operable during this event. Though the HPCI system did not meet the surveillance requirement response time, the system was available to perform its safety function of injecting the required flow rate into the Reactor Pressure Vessel.

Corrective Actions

The following corrective actions were completed.

- On June 15, 2019, the EG-R Actuator was replaced and the testing was repeated successfully with the response time acceptance criteria met.
- Historical data was reviewed on Unit 1 and it was confirmed that this condition was isolated to Unit 2. This action was complete on June 13, 2019.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

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1. FACILITY NAMEBrunswick Steam Electric Plant (BSEP),
Unit 2**2. DOCKET NUMBER****05000324****3. LER NUMBER**

YEAR	SEQUENTIAL NUMBER	REV NO.
2019	- 003	- 00

NARRATIVE

The following corrective actions are currently planned. Any changes to the corrective actions and schedules noted below will be made in accordance with the site's corrective action program.

- The failed EG-R Actuator was sent to a third-party vendor for post failure analysis. Once the vendor report is received, follow-up actions will be initiated as necessary. Initiation of necessary actions is scheduled to be complete by August 29, 2019.

Previous Similar Events

There have been no events in the past three years in which the HPCI System was made inoperable due to a failure with its governor system.

Commitments

No regulatory commitments are contained in this report.